

Module Review: Bacterial glutamate→P5C branch of proline biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP000000556) **Resolved bucket:** `kegg:ppu00332` ("Carbapenem biosynthesis") — see §5, this label is a precursor-overlap artifact. **Candidate genes:** proB (PP_0691 / Q88Q07), proA (PP_4811 / Q88DL4)

1. Executive summary

- **The module is SATISFIED.** Both expected steps of the glutamate→L-glutamate-5-semialdehyde/P5C branch are encoded in KT2440 by high-confidence, reviewed (Swiss-Prot) genes: **proB (PP_0691)** = glutamate 5-kinase (EC 2.7.2.11) and **proA (PP_4811)** = γ-glutamyl phosphate reductase (EC 1.2.1.41). Both are single-copy for their KEGG ortholog and map to UniPathway UPA00098 (L-proline biosynthesis), steps 1 and 2.
- **The resolved pathway name is misleading.** KEGG `ppu00332` "Carbapenem biosynthesis" contains **only** proB and proA; the carbapenem-committed enzymes (carA K21284, carB K21283, carC K21298) are **entirely absent** from KT2440. proB/proA appear in that map solely because pyrroline-5-carboxylate (P5C) is a shared precursor. The correct biological home is **proline biosynthesis** (KEGG map00330 / module M00015; UniPathway UPA00098). → Flag the bucket `module_needs_revision`.
- **The module boundary is correct for this organism.** KEGG module M00015 (`((K00931 K00147), K12657) K00286`) shows the bacterial two-gene branch (proB+proA) is the *sole* Glu→P5C route in KT2440 (no eukaryotic bifunctional P5CS, K12657, is present). The terminal P5C reductase **proC (PP_5095)** is present but is intentionally outside module scope.
- **Evidence type:** Direct KT2440 gene/genome evidence (UniProt Swiss-Prot rule-based annotation, KEGG genome). Enzyme mechanism/regulation is transferred from well-studied *E. coli* orthologs (strong transfer; conserved family, HAMAP rules). No direct KT2440 enzymology was found.

2. Target-organism pathway definition

Process included (module scope): the two committed cytoplasmic reactions converting L-glutamate to L-glutamate-5-semialdehyde (GSA), which non-enzymatically cyclizes to Δ^1 -pyrroline-5-carboxylate (P5C):

1. **L-glutamate + ATP → L-glutamyl-5-phosphate + ADP** (glutamate 5-kinase, ProB, EC 2.7.2.11; Rhea:14877)

2. **L-glutamyl-5-phosphate + NADPH + H⁺ → L-glutamate-5-semialdehyde + Pi + NADP⁺** (γ -glutamyl phosphate reductase, ProA, EC 1.2.1.41; Rhea:19541)

Deliberately excluded: the terminal reduction **P5C + NADPH → L-proline** (P5C reductase, ProC, EC 1.5.1.2), encoded by PP_5095 in KT2440.

Neighboring pathways/maps to keep SEPARATE: - **Proline catabolism (utilization):** bifunctional **PutA (PP_4947, K13821)** oxidizes proline → P5C → glutamate — the *reverse/degradative* direction; not part of biosynthesis. - **Ornithine→proline bypass:** ornithine cyclodeaminase **ocd (PP_3533, K01750)** makes proline from ornithine *without* traversing Glu→P5C. - **Arginine biosynthesis:** N-acetylglutamate kinase **argB (PP_5289, K00930)** and N-acetyl- γ -glutamyl-phosphate reductase **argC (PP_0432, PP_3633, K00145)** are the same-superfamily paralogs of proB/proA (see §5). - **Broad overview maps** (ppu01100 Metabolic pathways, ppu01110 Biosynthesis of secondary metabolites, ppu01230 Biosynthesis of amino acids) contain proB/proA but should not define the module. - **Carbapenem biosynthesis (ppu00332):** not a real pathway in KT2440 (§5).

Alternate names / database definitions: ProB = glutamate 5-kinase = γ -glutamyl kinase (GK); ProA = γ -glutamyl phosphate reductase (GPR) = glutamate-5-semialdehyde dehydrogenase (GSA dehydrogenase). UniPathway UPA00098 "L-proline biosynthesis"; KEGG module M00015 "Proline biosynthesis, glutamate => proline"; MetaCyc PROSYN-PWY.

Genomic organization (species-specific): In KT2440 the three pro genes are **unlinked** — proB PP_0691 at ~0.80 Mb (+ strand), proA PP_4811 at ~5.47 Mb (– strand), proC PP_5095 at ~5.82 Mb (+ strand) — i.e., proB and proA are ~4.67 Mb apart on opposite strands. This differs from the co-transcribed **proBA operon** of *E. coli/Serratia* (

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The ProB→ProA metabolic relationship holds, but operon-/synteny-based co-regulation or joint annotation transfer must **not** be assumed for this taxon; each gene stands on its own evidence.

3. Expected step model

#	Step	Enzyme (family)	KO	EC	KT2440 gene	Status
1	Glutamate phosphorylation	Glutamate 5-kinase / ProB (amino-acid-kinase AAK family + PUA domain)	K00931	2.7.2.11	PP_0691 (Q88Q07)	covered
2	γ -Glutamyl phosphate reduction $\rightarrow$ GSA/P5C	γ -Glutamyl phosphate reductase / ProA (aldehyde-dehydrogenase family)	K00147	1.2.1.41	PP_4811 (Q88DL4)	covered
(3)	P5C $\rightarrow$ L-proline (terminal, out of scope)	P5C reductase / ProC	K00286	1.5.1.2	PP_5095	present (excluded from module)

No expected in-scope step is missing. No lineage-specific replacement of steps 1–2 exists (no bifunctional P5CS K12657).

4. Candidate genes and evidence

proB — PP_0691 (UniProt Q88Q07, PROB_PSEPK) — HIGH confidence

- **Role:** Glutamate 5-kinase (EC 2.7.2.11), committed first step; catalyzes L-glutamate + ATP $\rightarrow$ L-glutamyl-5-phosphate (Rhea:14877). 372 aa.
- **Evidence:** Reviewed Swiss-Prot; annotation via HAMAP rule MF_00456 (PE=3, inferred from homology). Single gene for K00931 in KT2440; UniPathway UPA00098:UER00359. RefSeq WP_003247458.1; BioCyc PPUT160488 monomer.

- **Domain architecture:** AAK catalytic domain (Pfam PF00696) + C-terminal PUA regulatory domain (Pfam PF01472). In *E. coli* the AAK domain performs catalysis and proline feedback inhibition; the PUA domain modulates Mg^{2+} requirement and proline-triggered oligomerization

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Strong transfer to KT2440 (conserved domain set).

- **Curation caveat:** activity is inferred (rule-based), not measured in KT2440. The AAK fold is shared with arginine-pathway argB (NAGK); however KO and domain context resolve them cleanly. Proline feedback inhibition is expected but unverified in KT2440 (relevant if the module is used for flux/metabolic-engineering reasoning).

proA — PP_4811 (UniProt Q88DL4, PROA_PSEPK) — HIGH confidence

- **Role:** γ -Glutamyl phosphate reductase (EC 1.2.1.41); NADPH-dependent reduction of L-glutamyl-5-phosphate $\rightarrow$ L-glutamate-5-semialdehyde (Rhea:19541). 423 aa.
- **Evidence:** Reviewed Swiss-Prot; HAMAP rule MF_00412 (PE=3). Single gene for K00147; UniPathway UPA00098:UER00360. RefSeq WP_004576854.1.
- **Domain architecture:** aldehyde-dehydrogenase domain (Pfam PF00171); GPR-specific InterPro IPR000965/IPR012134.
- **Curation caveat:** shares the aldehyde-DH fold with arginine-pathway argC; KO/InterPro resolve them. Activity inferred, not directly measured in KT2440.

GO annotations (both, IEA/UniRule): proB GO:0004349 (glutamate 5-kinase activity); proA GO:0004350 (glutamate-5-semialdehyde dehydrogenase activity); both GO:0055129 (L-proline biosynthetic process). proB also carries GO:0003723 (RNA binding, from the PUA domain via InterPro) — a **broad/likely-spurious** transfer for a biosynthetic kinase; treat with caution.

5. Gaps, ambiguities, and likely over-annotations

1. **Bucket mislabel (highest priority).** kegg:ppu00332 "Carbapenem biosynthesis" is **not a real pathway in KT2440**: it contains only proB/proA, and all carbapenem-committed genes (carA/carB/carC) are absent. This is a KEGG map artifact driven by the shared P5C precursor. $\rightarrow$ **module_needs_revision**; re-anchor the module under proline biosynthesis

(map00330 / M00015 / UPA00098). Carbapenem-specific steps →
not_expected_in_target_taxon.

2. **Paralog ambiguity (low actual risk).** proB↔argB (PP_5289; AAK family) and proA↔argC (PP_0432, PP_3633; aldehyde-DH family) are homologous, but are cleanly resolved by **distinct EC numbers and KOs**: argB = acetylglutamate kinase EC 2.7.2.8 (K00930) vs proB EC 2.7.2.11 (K00931); argC = N-acetyl-γ-glutamyl-phosphate reductase EC 1.2.1.38 (K00145) vs proA EC 1.2.1.41 (K00147). Since they differ at the 4th EC digit and carry different KOs (plus proB's diagnostic PUA domain), EC/KO-based curation will not cross-assign them — over-propagation risk is genuinely low. (Nuance: KT2440 has two argC-family copies, PP_0432 and PP_3633 [latter "putative"], a duplication within the *arginine* branch, irrelevant to this module.)
 3. **Broad GO term.** proB's GO:0003723 (RNA binding) is a PUA-domain InterPro side-effect, not a demonstrated function; potentially over-propagated.
 4. **Direction/utilization confusion.** PP_4947 (PutA) and PP_3533 (ocd) are proline *catabolism* and an *ornithine bypass*; they must not be counted toward this biosynthetic module.
 5. **No direct enzymology.** All KT2440 evidence is rule/homology-based (PE=3); no target-strain kinetic or knockout data was located. This is the main evidence gap, though transfer confidence is high given deep conservation.
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6. Module and GO-curation recommendations

Item	Recommendation
Step 1 (ProB glutamate 5-kinase)	covered — PP_0691
Step 2 (ProA γ -glutamyl phosphate reductase)	covered — PP_4811
Terminal P5C reductase (proC)	out of module scope; present as PP_5095 (note for completeness)
Carbapenem-committed steps	not_expected_in_target_taxon (carA/carB/carC absent)
Bucket <code>kegg:ppu00332</code>	module_needs_revision — relabel/re-anchor to proline biosynthesis (map00330 / M00015 / UPA00098); "Carbapenem biosynthesis" is a shared-precursor artifact
Module boundaries	correct as written (Glu→GSA/P5C, stopping before ProC); matches the bacterial <code>(K00931 K00147)</code> arm of M00015
New GO term requests	none needed; existing GO:0004349, GO:0004350, GO:0055129 are adequate. Consider deprecating/flagging proB GO:0003723 (RNA binding) as likely over-propagated

No new module document is required beyond correcting the pathway anchor. If the curation system tracks the terminal step, a sibling module (P5C→proline, ProC/PP_5095) could be added, but it is out of the present module's scope.

7. Genes to promote to full `fetch-gene` review

- **PP_0691 (proB)** — promote. Rationale: it is the committed, feedback-regulated control point; confirm domain architecture (AAK+PUA), single-copy status, and that GO:0003723 is not carried forward. Low ambiguity but high pathway importance.
- **PP_4811 (proA)** — promote. Rationale: confirm EC 1.2.1.41 assignment vs. argC paralogy; single-copy status.

- **Context genes (review only if module is extended):** PP_5095 (proC, terminal step), PP_5289 (argB paralog), PP_0432/PP_3633 (argC paralogs), PP_4947 (PutA, catabolism), PP_3533 (ocd). These are *not* part of this module but are relevant to keep boundaries clean.

8. Key references

- Marco-Marín C, et al. *Crystal structure of E. coli glutamate 5-kinase: a novel two-domain (AAK+PUA) architecture; feedback-inhibited by proline/ornithine*. J Mol Biol. 2007.

P 17321544

- Pérez-Arellano I, Rubio V, Cervera J. *Mapping active-site residues in glutamate-5-kinase: glutamate and proline bind at overlapping sites*. FEBS J. 2006.

P 17069808

- Pérez-Arellano I, Rubio V, Cervera J. *Dissection of E. coli glutamate 5-kinase: functional impact of PUA-domain deletion*. FEBS Lett. 2005.

P 16337196

- Omori K, et al. *Serratia marcescens proBA operon: proB (GK) and proA (GPR) organization and proline feedback control*. J Bacteriol. 1991 / Mol Gen Genet 1992.

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P 1316937

- Fujita T, et al. *Allosteric regulation of γ -glutamyl kinase by proline (tomato tomPRO1)*. Plant Physiol. 2003.

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- Database resources (direct KT2440 evidence): UniProt **Q88Q07** (PROB_PSEPK), **Q88DL4** (PROA_PSEPK); KEGG pathway **ppu00332**, module **M00015**; UniPathway **UPA00098** (UER00359/UER00360/UER00361); KEGG orthology K00931/K00147/K00286/K00930/K00145/K13821/K01750.

Uncertainty & species-transfer notes

- **Direct (KT2440):** gene presence, single-copy KO mapping, EC/pathway assignment, absence of carbapenem genes, presence of proC/PutA/ocd/argB/argC — all from KT2440 genome/UniProt records.
- **Transferred (strong):** enzyme mechanism, domain function, proline feedback inhibition — from *E. coli* and other bacteria; family is deeply conserved and covered by HAMAP rules, so transfer to KT2440 is strong though not experimentally confirmed in-strain.
- **Open questions:** Is ProB feedback inhibition operative/measurable in KT2440 (relevant for proline/osmolyte engineering)? Any in-strain knockout/kinetic data? These would elevate evidence from homology-based to direct.

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